

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459007.7595 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.009 +/- 0.053
Transit depth (Rp/Rs)^2	0.0084 +/- 0.096 [%]
Orbital Inclination (inc)	84.65 +/- 1.83
Airmass coefficient 1 (a1)	1.7 +/- 0.88
Airmass coefficient 2 (a2)	-0.48 +/- 1.31
Scatter in the residuals of the lightcurve fit is	52.78 %
Best Comparison Star	None
Optimal Aperture	2.92
Optimal Annulus	5.63
Transit Duration (day)	0.014 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.009 ± 0.053 vs 0.1326 (deviation 2.33σ)
Duration fit vs published	0.014 d vs 0.1238 d (deviation 6.1σ)
Mid-time O–C	-4.6 min
Depth SNR	0.2
Residual scatter	52.78 %

Light curve

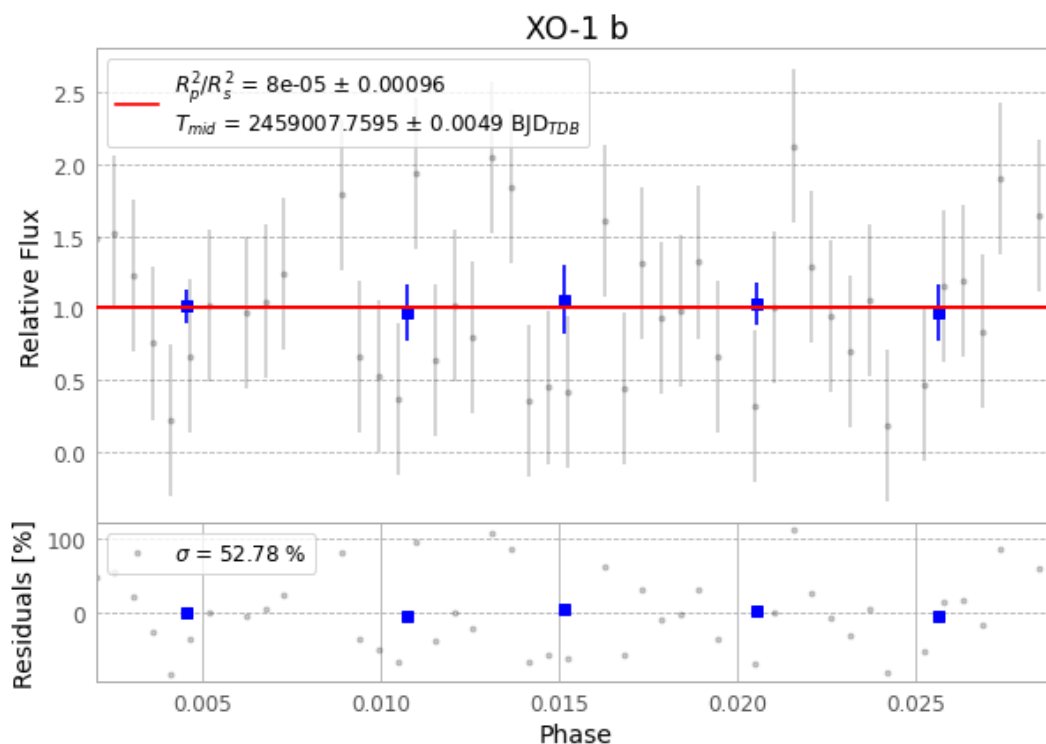


Figure 1 — FinalLightCurve_XO-1 b_2020-06-06.png (SHA-256 d8151d42fb5bcaeb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [223, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0e1a08feb4545c5b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

52 FITS frames (34 MB), XO-1200607061815.FITS → XO-1200607085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-1-200607.log (e3584370eb7d...), FinalLightCurve_XO-1 b_2020-06-06.png (d8151d42fb5b...), FinalLightCurve_XO-1 b_2020-06-06.csv (bd51d237c3cc...)

Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:17Z.